Vision Point Software
Installation Guide

December 2025
Rev 6.9.0



20283 State Road 7, Suite 350
Boca Raton, FL 33498
USA
+1 (561) 689-2899

info@kayainstruments.com
support@kayainstrunments.com

www.kayainstruments.com

+1 (561) 689-2899
info@kayainstruments.com
www.kayainstruments.com

Revision History

Version	Date	Notes
1.0	08/2014	Initial release
1.1	09/2014	Predator API release 1.0.5.1 - Fix parameter types signature. - Additional working examples.
1.0	08/2014	Predator APP release 1.0.0 - Initial release
2.0	10/2015	Predator APP release 2.0.0 - Minor changes
3.0	03/2016	Vision Point APP release 3.0.0 - The product line "Predator" has been renamed "Vision Point".
4.0	11/2016	Vision Point APP release 4.0.0 - Installation for Ubuntu was added
4.1	07/2017	Vision Point APP release 4.1 - Documentation improvements
4.2	09/2017	Vision Point APP release 4.2 - New installation notes section
4.3	04/2018	Vision Point APP release 4.3 - Added contact and support information section
4.4	09/2018	Vision Point APP release 4.4 - Documentation improvements
5.0	03/2019	Vision Point APP release 2019.1/API 5.0 - Additional installation steps and instructions were added for Windows OS users - Documentation improvements
5.0.1	05/2019	Vision Point APP release 2019.1/API 5.0.1 - Review and minor documentation corrections
5.1	07/2019	Vision Point APP release 2019.2/API 5.1 - Review and minor corrections for Windows users
5.1	12/2019	Vision Point APP release 2019.2/API 5.1 - Review and minor corrections in system requirements sections
5.2	06/2020	Vision Point APP release 2020.1/API 5.2 - Windows 7 is no longer supported - Document rearrangement
5.3	08/2020	Vision Point APP release 2020.2/API 5.3 - System requirements update - Minor corrections in installation for Linux section
5.4	12/2020	Vision Point APP release 2020.3/API 5.4 - Review and minor corrections
5.4.1	04/2021	Vision Point APP release 2020.3/API 5.4 (Service pack 1) - Review and minor corrections
6.0	09/2021	Vision Point APP release 2021.1/API 6.0 - Installation instruction for CentOS and RedHat section was removed - Review and minor corrections
6.0.1	01/2022	Vision Point APP release 2021.1/API 6.0.1 - Added Nvidia Jetson Xavier system requirements
6.1	04/2022	Vision Point APP release 2022.1/API 6.1 - Python examples were added to installation package to support multiple camera flow - “KYInfo.bat” file was added to automate collection of diagnostic info

		- Added DKMS driver signature section
6.2	09/2022	Vision Point APP release 2022.2/API 6.2 - Added Ubuntu instllation procedure flags - Added Ubuntu uninstallation procedure section
6.3.0	01/2023	Vision Point APP release 2023.1/API 6.3.0 - Updated system requirements section - Added application installation administrator privileges requirement
6.3.10	02/2024	Vision Point APP release 2023.1/API 6.3.10 - Ubuntu 22.04 kernel driver update - New device support: Predator II Single CoaXPress M.2
6.4.0	04/2024	Installation Procedure for Windows section updated
6.4.1 - 6.9.0	10/2024-12/2025	The same document without changes

Table 1 – Revision History

+1 (561) 689-2899
info@kayainstruments.com
www.kayainstruments.com

Table of Contents

Revision History 1
1 Figures and Tables 4
1.1 List of Figures 4
1.2 List of Tables 4
2 Introduction 5
2.1 Safety precautions 5
2.2 Disclaimer 6
2.3 Important Notes 6
3 Installation Procedure for Windows 7
3.1 System Requirements 7
3.2 Installation Procedure 8
4 Installation Procedure for Ubuntu 14
4.1 System Requirements 14
4.1.1 Ubuntu 20.04 / 22.04 14
4.1.2 Nvidia Jetson Xavier 14
4.2 Installation Procedure 15
4.3 Uninstallation Procedure 16
4.4 Signing hardware driver using DKMS 17
4.5 Building hardware driver 20
5 Installation Notes 21
5.1 How to downgrade Vision Point application from version 2019.1 to 4.4 21
5.2 Driver installation for Windows 22
5.3 Software and firmware upgrade for Windows 23

1 Figures and Tables

1.1 List of Figures

Figure 1 – Vision Point Welcome screen 8
Figure 2 – Vision Point installation location 8
Figure 3 – Installation components checkbox 9
Figure 4 – Start Menu shortcut folder 9
Figure 5 – Pre-installation review 10
Figure 6 – Installation process bar 10
Figure 7 – Back up your work 11
Figure 8 – Device driver wizard access dialog 11
Figure 9 – Device driver installation wizard 12
Figure 10 – Completing the device driver installation wizard for Windows OS 12
Figure 11 – Completing the installation 13
Figure 12 – DKMS user input 17
Figure 13 – Secure boot configuration message 17
Figure 14 – Secure boot configuration password 18
Figure 15 – Images ‘a’ to ‘e’: Enrolling MOK instructions 19

1.2 List of Tables

Table 1 – Revision History 2

2 Introduction

2.1 Safety precautions

With your KAYA's board in hand, please take the time to read through the precautions listed below to prevent preventable and unnecessary injuries and damage to you, other personnel, or property. Read these safety instructions carefully before your first use of the product, as these precautions contain safety instructions that must be observed. Be sure to follow this manual to prevent misuse of the product.

⚠ Caution! Read Carefully and do not disregard these instructions.
In case of a failure, disconnect the power supply Disconnect the power supply immediately and contact our sales personnel for repair. Continuing to use the product in this state may result in a fire or electric shock.
If an unpleasant smell or smoking occurs, disconnect the power supply. Disconnect the power supply immediately! Continuing to use the product in this state may result in a fire or electric shock. After verifying that no smoking is observed, contact our sales personnel for repair.
Do not disassemble, repair or modify the product. It may result in a fire or electric shock due to a circuit shortage or heat generation. Contact our sales personnel before inspection, modification, or repair.
Do not place the product on unstable surfaces. Otherwise, it may drop or fall, resulting in injury to persons or the camera.
Do not use the product if dropped or damaged. Otherwise, a fire or electric shock may occur.
Do not touch the product with metallic objects. Otherwise, a fire or electric shock may occur.
Do not place the product in dusty or humid environments, nor where water may splash. Otherwise, a fire or electric shock may occur.
Do not wet the product or touch it with wet hands. Otherwise, the product may fail or cause a fire, smoking, or electric shock.
Do not touch the gold-plated sections of the connectors on the product. Otherwise, the surface of the connector may be contaminated by sweat or skin oil, resulting in contact failure of a connector, malfunction, fire, or electric shock due to static electricity discharge.
Do not use or place the product in the following locations. ▪ Unventilated areas such as closets or bookshelves. ▪ Near oils, smoke, or steam. ▪ Next to heat sources. ▪ A closed (and not running) car where the temperature becomes high. ▪ Static electricity replete locations ▪ Near water or chemicals. Otherwise, a fire, electric shock, accident, or deformation may occur due to a short circuit or heat generation.
Do not place heavy objects on the product. Otherwise, the product may be damaged.
Be sure to discharge static electricity from the body before touching any sensitive electronic components. The electronic circuits in your computer and the circuits on the ***Iron*** camera and the ***Predator II*** board are sensitive to static electricity and surges. Improper handling may seriously damage the circuits. In addition, do not let your clothing come in contact with the circuit boards or components. Otherwise, the product may be damaged.

2.2 Disclaimer

KAYA Instruments assumes no responsibility for any damage that may ensue by using this product for any purpose other than intended, as previously stated. Without detracting from what was previously written, the company takes no responsibility for any damages caused by:

- Earthquake, thunder strike, natural disasters, a fire caused by use beyond our control, willful and/or accidental misuse and/or use under other abnormal and/or unreasonable conditions.
- Secondary damages caused by the use of this product or its unusable state (business interruption or others).
- Use of this product in any manner that contradicts this manual or malfunctions due to connection to other devices. Damage to this product that is out of our control or failure due to modification
- Accidents and/or third parties that may be involved.

Additionally, KAYA Instruments assumes no responsibility or liability for:

- Erasure or corruption of data caused by the use of this product.
- Any consequences or other abnormalities following the use of this product

2.3 Important Notes



Important notes

1. Vision Point application requires **administrator** privileges. Please make sure this requirement is met prior to installation execution.
2. The Iron IMX camera's firmware **must** be upgraded to version 4.0 (or higher) **before** installing Vision Point 2021.1 software. The latest firmware can be found here.

+1 (561) 689-2899
info@kayainstruments.com
www.kayainstruments.com

3 Installation Procedure for Windows

3.1 System Requirements

Before installing, please, make sure your system meets the following requirements:

- Intel x64 processor or compatible
- At least 4 GB of system memory
- Windows 10 64-bit, Windows 11-64 bit operating systems
- Hard drive with 400 MB of free space
- At least one of KAYA Instruments boards installed
- Internet connection

Remarks:

1. Vision Point 2019.1 (API 5.1) is the latest version to support Windows 7. We encourage our customers to switch to the supported Windows OS to support our latest updates and hotfixes.
2. Vision Point 2020.3 is the last version to support Windows 10 32-bit OS.
3. For Windows OS to support the latest version of Vision Point, please make sure your Windows is up to date, and all the latest updates and hotfixes are installed.
   a. If your computer hardware does not support Windows 7 latest updates, please consider using the Vision Point 4.4 software version.
   b. To downgrade the Vision Point application from version 2019.1 to 4.4, please refer to the section 5.1.
4. Genicam and OpenCV libraries will no longer be installed nor registered in the user directory path. During application installation, those are installed for internal use only and won't be accessible for the user, and thus if needed by the user's application, it should be installed separately.
5. Visual Studio 2017 flavor should be used on run-time. Visual Studio 2012 run-time is no longer supported nor included in the installation.
6. Inserting and/or removing KAYA PCI devices requires a reboot of the computer or restart of "KAYA Instruments" service. After that, one may use Vision Point Application or open API examples with KAYA devices.

3.2 Installation Procedure

1. Within the Explorer window, open the folder that contains the installation file: KAYA_Vision_Point_Setup.exe and execute with administrator privileges.
2. After installing Utility, follow the instructions as described below.
3. On the Welcome screen, click "Next".

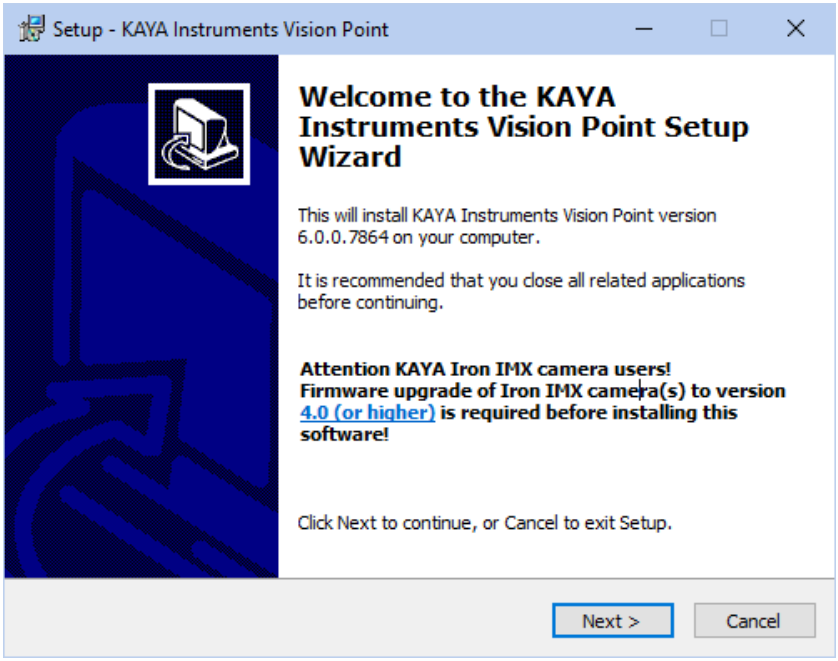


Figure 1 – Vision Point Welcome screen

4. At this stage, you will have to define the target folder for the installation. It is recommended to keep the default folder. After you select the installation folder, click the "Next" button.

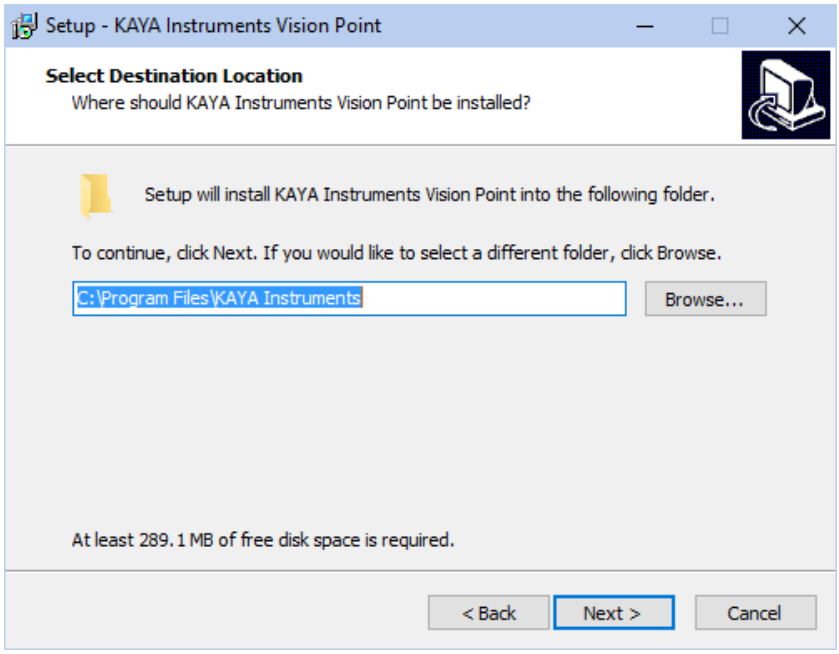


Figure 2 – Vision Point installation location

5. Install serial communication Virtual COM Port driver. The interface is available only for supported Frame Grabber devices.

   NOTE: *Serial communication interface, which doesn't require the Virtual COM Port, will still be available for supported devices.*

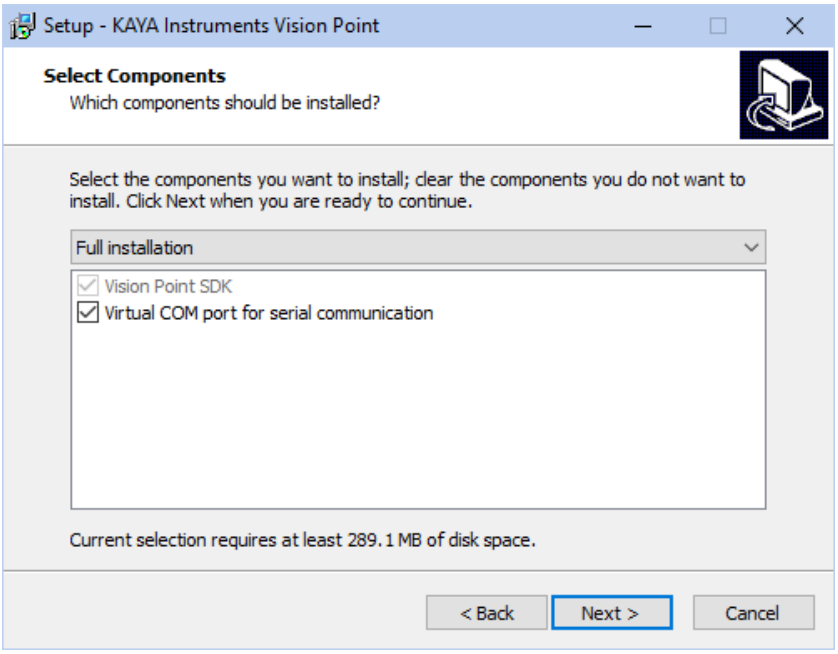


Figure 3 – Installation components checkbox

6. Now you will have to select the Start Menu folder for the software installation. It is recommended to keep the default location. Click the "Next" button to proceed.

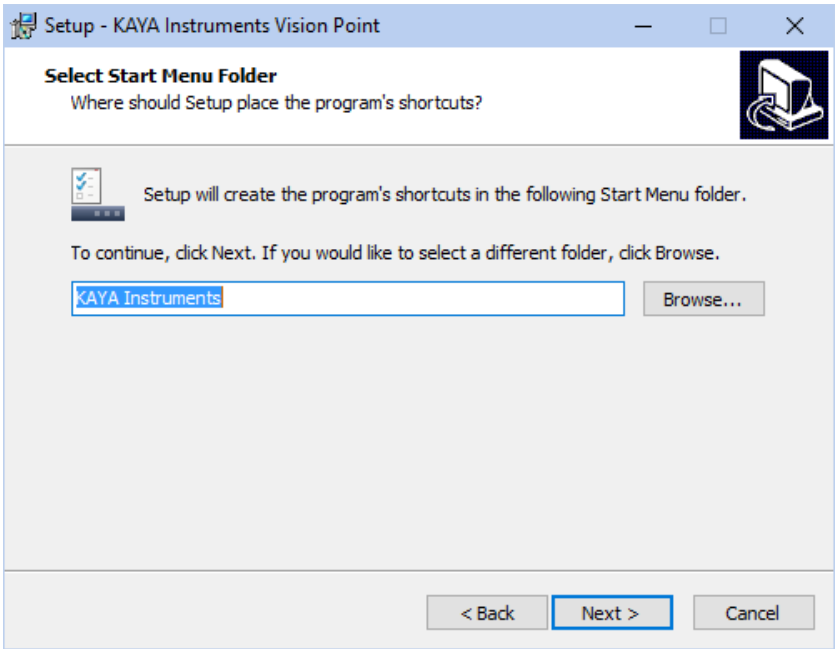


Figure 4 – Start Menu shortcut folder

7. At this stage, you will have the chance to review the settings before actual software installation. After you click "Install", the installation procedure will start. It will take a few minutes for the installation to complete. A few popup windows will appear during the installation.

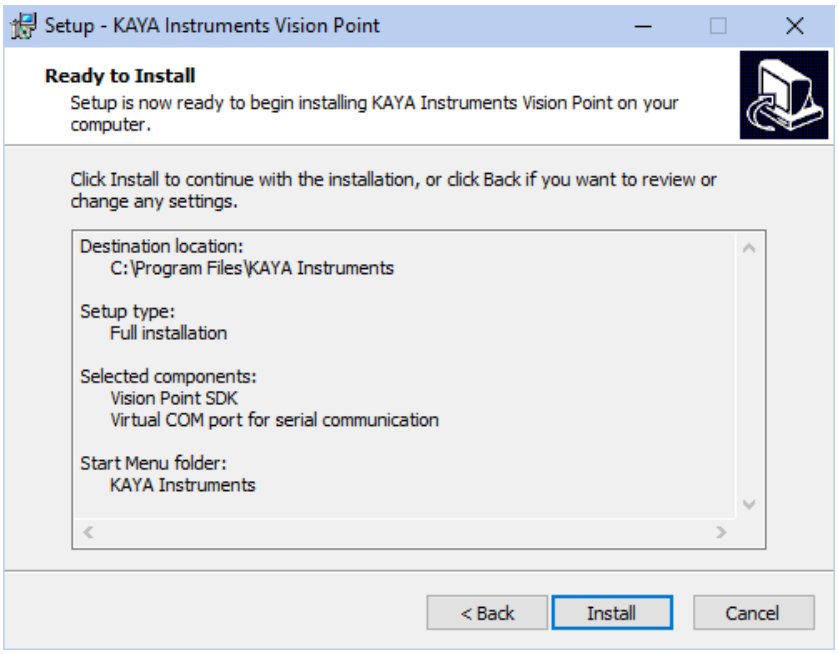


Figure 5 – Pre-installation review

8. During the installation procedure, a few popup windows will appear. Please follow the instructions listed in the next installation steps.

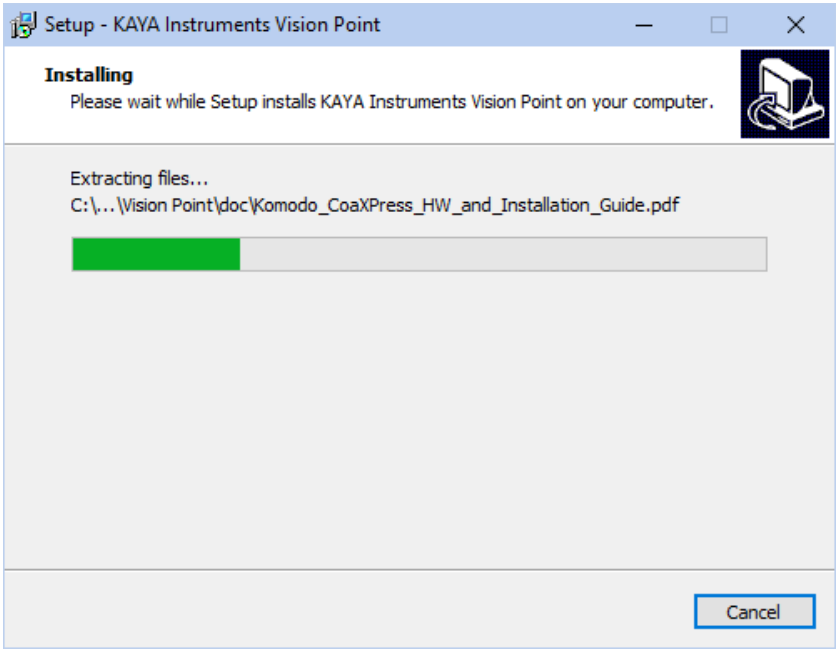


Figure 6 – Installation process bar

9. Please back up your work and example modifications if the Vision Point application was previously installed on your computer.

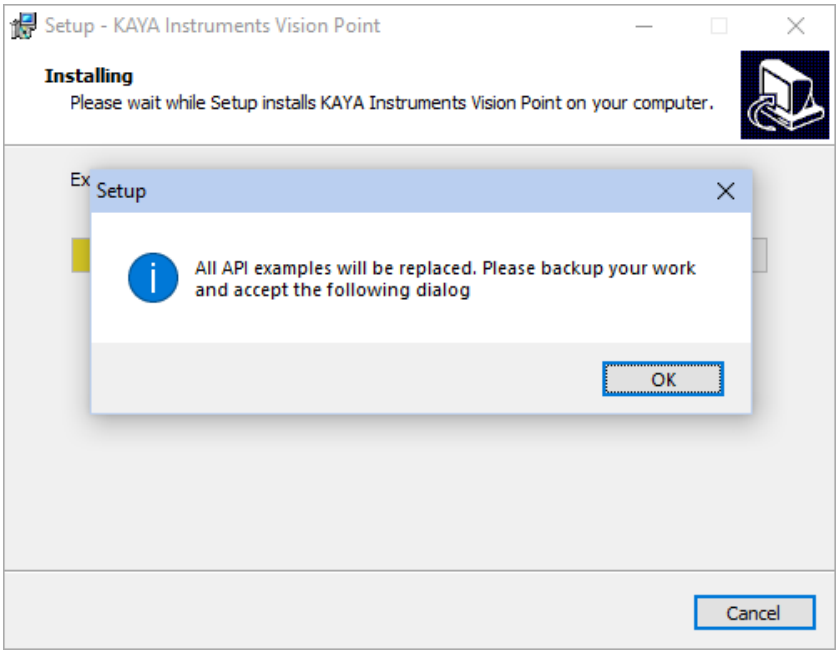


Figure 7 – Back up your work

10. Please accept the following dialog to access the driver installation wizard.

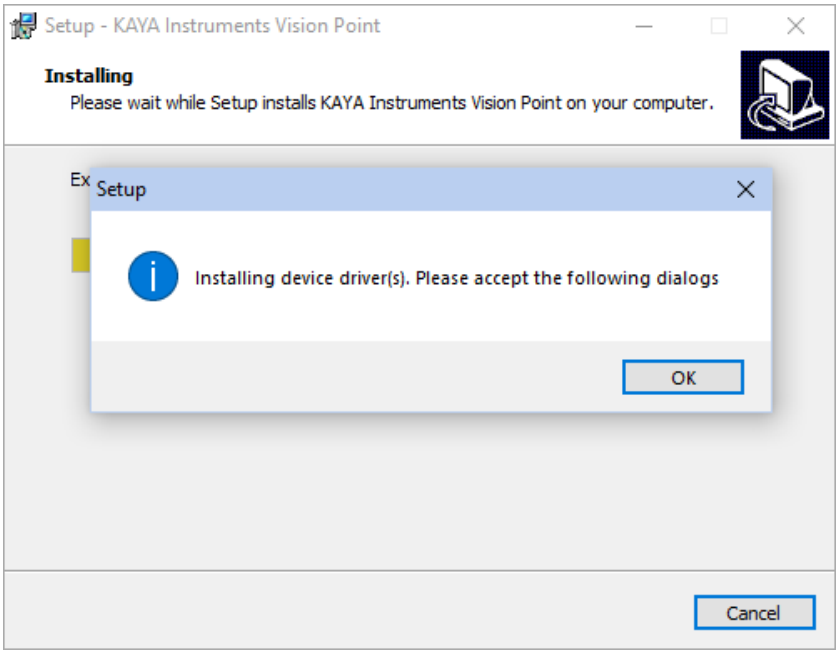


Figure 8 – Device driver wizard access dialog

11. Click the "Next" button to proceed to the device driver installation wizard.

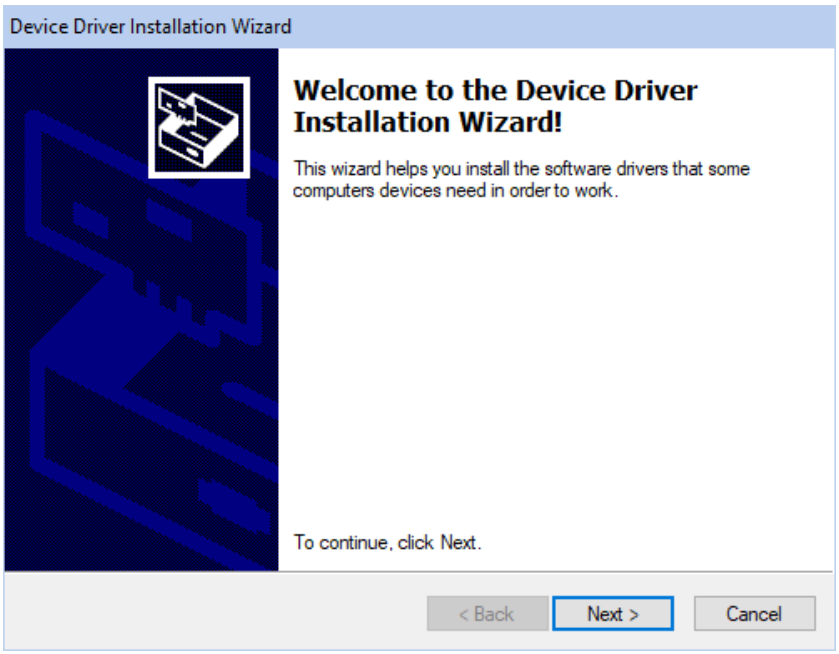


Figure 9 – Device driver installation wizard

12. Click "Finish" to finalize the device driver installation wizard.

NOTE: *After pressing "Finish", a few popup windows will appear and disappear immediately.*

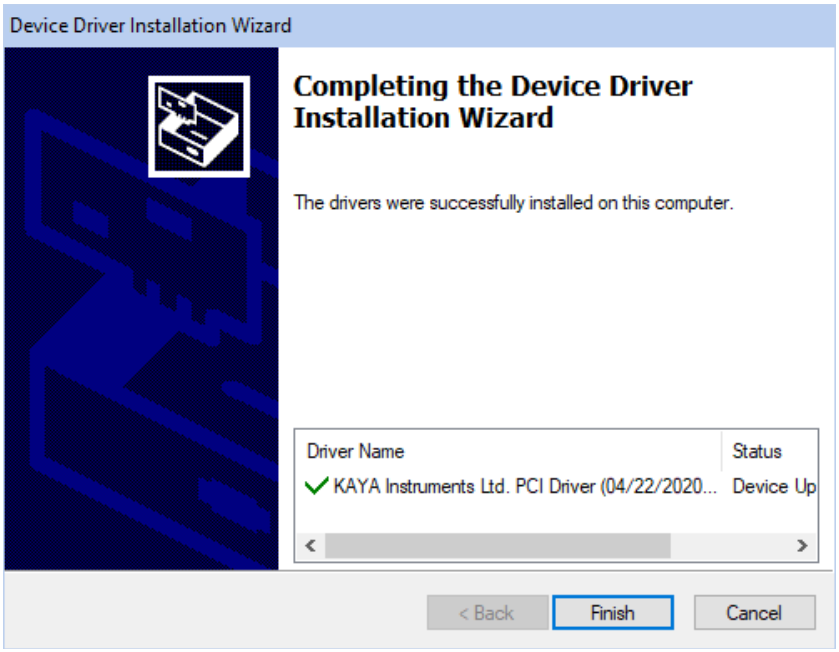


Figure 10 – Completing the device driver installation wizard for Windows OS

13. Windows OS requires a reboot of the computer in order to complete the installation process. After a reboot, you are welcome to start Vision Point Application or open API examples.

    New documentation was installed during the installation process. We encourage you to read it and familiarize yourself with new and improved features. Open the documentation folder using the link in the installation window.

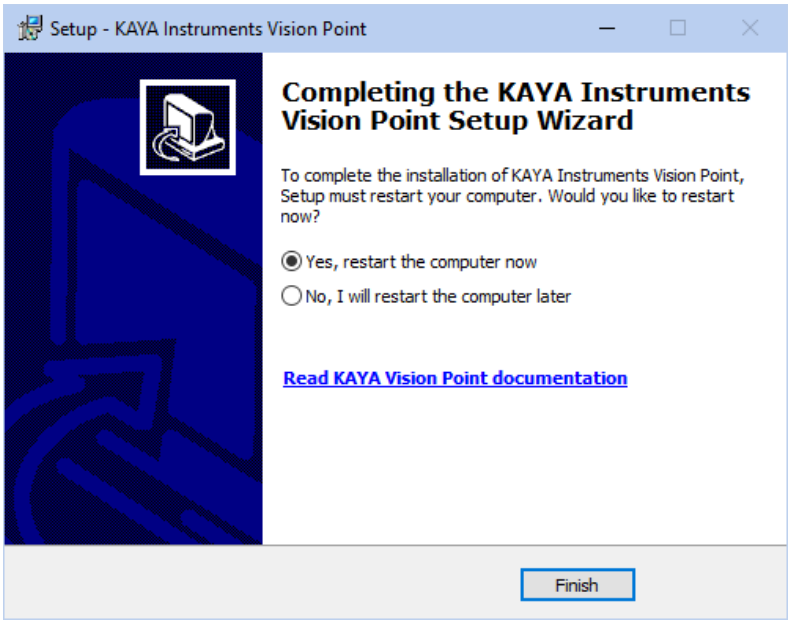


Figure 11 – Completing the installation

Installation log: The Vision Point application installation log files folder can be found under user's main driver: C:\Program Files\KAYA Instruments\Log\Installer folder.

4 Installation Procedure for Ubuntu

4.1 System Requirements

Before installing, please make sure your system meets the following requirements:

4.1.1 Ubuntu 20.04 / 22.04

1. Ubuntu 20.04 / 22.04 64-bit Operating System installed.
2. 4 GB of RAM or more.
3. 400 MB available disk space for a default install.

4.1.2 Nvidia Jetson Xavier

1. Nvidia Jetson AGX Xavier module or development kit.
2. JetPack 5.0.2 or higher.
3. 400 MB available disk space for a default install.

IMPORTANT NOTES:

1. In case you are using secure boot please read section 4.4 before continuing to installation procedure.
2. The installation package contains an .sh launcher to launch the executable in the adjusted environment. Users are required to use the installed Vision Point launcher, or "VisionPoint.sh" in the worst case, and NOT to run the executable directly.
3. Starting from version 2023.1 (SP 6.3.9), Ubuntu 18.04 is no longer supported.

4.2 Installation Procedure

1. Extract the provided .tar.gz file using the following terminal command:

```
tar -zxvf KAYA_Vision_Point_Setup_2021.1_Ubuntu_20.04_x64.tar.gz
```

NOTE: *Installation archive name may vary*

2. Enter the "KAYA_Vision_Point" folder from the terminal using:

```
cd KAYA_Vision_Point
```

NOTE: *"KAYA_Vision_Point_Setup" may contain a suffix specifying OS name, version, architecture, etc., for example – "KAYA_Vision_Point_Setup_2021.1_Ubuntu_20.04_x64"*

3. Run the installation script with the following command:

```
sudo sh install.sh
```

This will install the Vision Point software and all required components and drivers for supported hardware

NOTE: *The installation package includes hardware drivers for several different Linux kernel versions and will try to select one that corresponds to your currently running kernel. If a version for your current kernel is not yet included in the package, you will see a message: "No suitable pre-built driver was found for your current kernel ..." Please refer to the section 4.5 "Building hardware driver".*
The following installation flags are available:

flag	Description
-s or -silent	Silent installation, no user input required
-n or -no_dkms	Use regular driver (without DKMS) installation methode
-d or -dkms	Use DKMS driver installation methode (avoid DKMS question)
-a or -no_alerts	Suppress all alerts and messages
--keep_driver	Keep current kernel driver. By default kernel driver is reinstalled by this script
--keep_daemon	Keep current service (daemon). By default service executable is replaced
--keep_tray	Keep current tray configuration. By default tray executable is replaced and added to autostart
--keep_conf	Keep current internal configuration. By default only public conf is retained and internal one is cleared
--help	Display the list of available flags

NOTE: *Type '--help' at the beginning of the installation to view the list of available flags.*

4. Reboot machine.
5. Vision Point Application can be found in the subfolder "/opt/KAYA_Instruments/bin", and the launch icon is placed on the desktop.
6. API usage sample can be found in subfolder "/opt/KAYA_Instruments/Examples".

4.3 Uninstallation Procedure

1. Enter the uninstallation folder from the terminal using:

```
cd /opt/KAYA_Instruments/lib
```

2. Run the installation script with the following command:

```
sudo sh uninstall.sh
```

The following installation flags are available:

flag	Description
-s or --silent	Silent uninstallation, no user input required

4.4 Signing hardware driver using DKMS

In Linux operating systems a secure boot process allows only approved drivers to run and requires hardware driver signature. To support this feature, the driver installation method has been changed, allowing the user to choose to add KAYA's driver to DKMS. The following steps explain the process of DKMS driver signing.

1. User may check whether a secure boot is activated and enabled for the OS using the following utility command:

```
mokutil --sb-state
```

NOTE: *In case the secure boot was not initially installed, this utility will not be present.*

2. Initiate the installation procedure, described in section 4.2.
3. During the installation procedure user input is required for choosing between default driver installation and signed driver installation using DKMS.

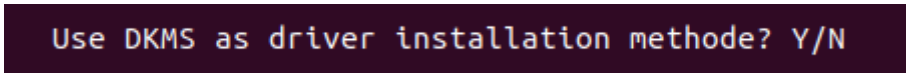


Figure 12 – DKMS user input

4. If secure boot is enabled, DKMS will try to sign the driver and the following message will pop up, in case systems MOK (Machine Owner Key) is NOT enrolled.

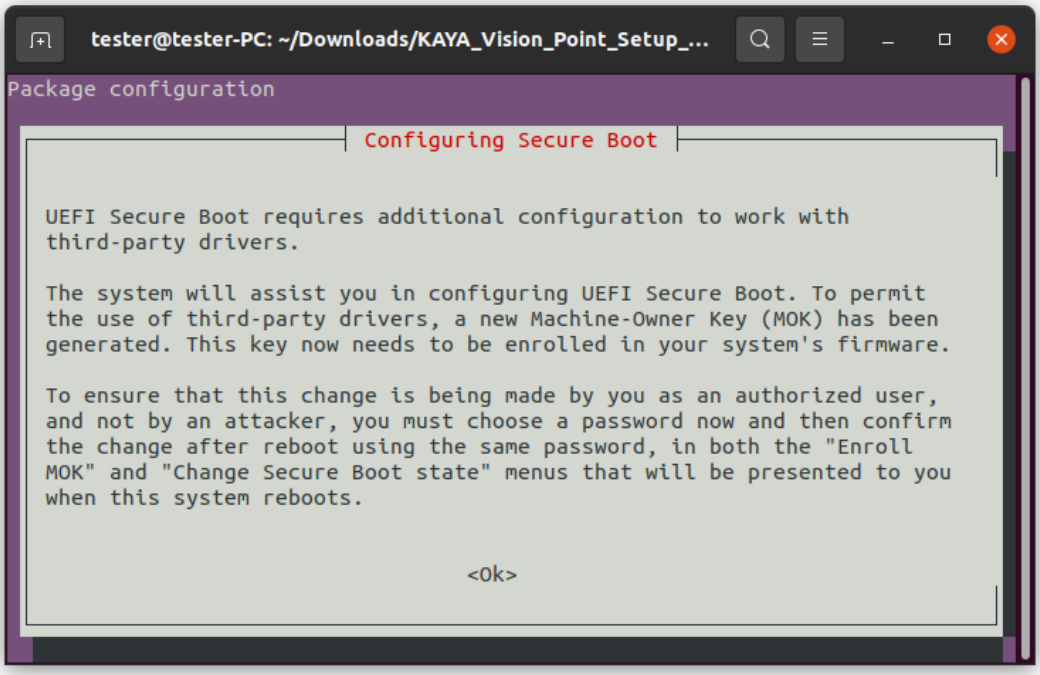


Figure 13 – Secure boot configuration message

5. Read the message and press "Ok".
6. Create the password. This is a one-time password, meaning it will be required later.

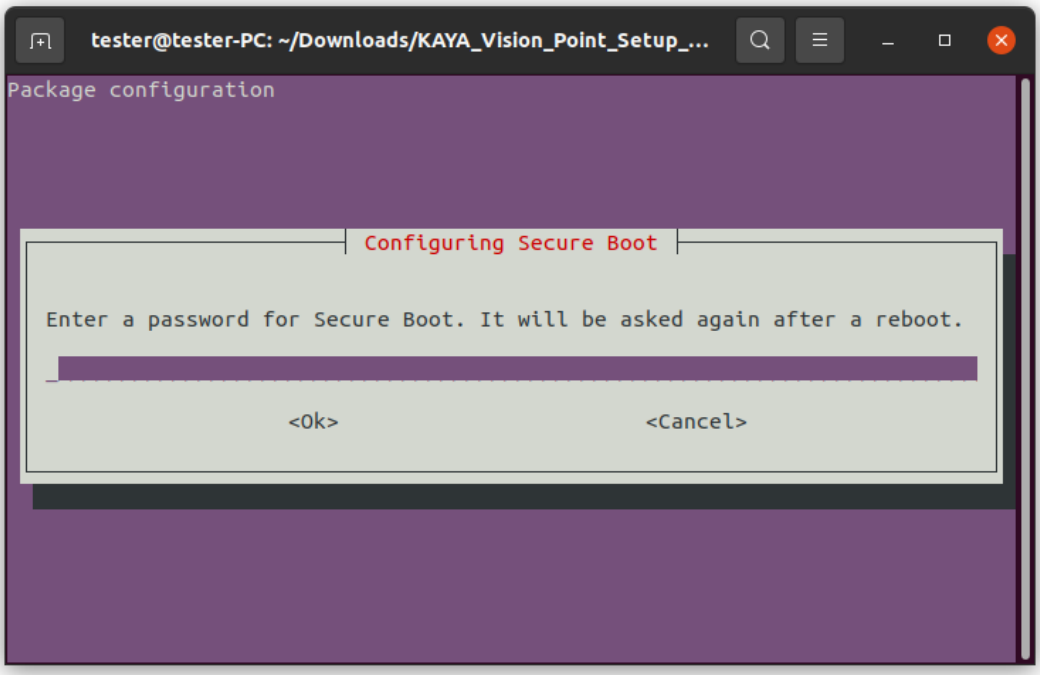
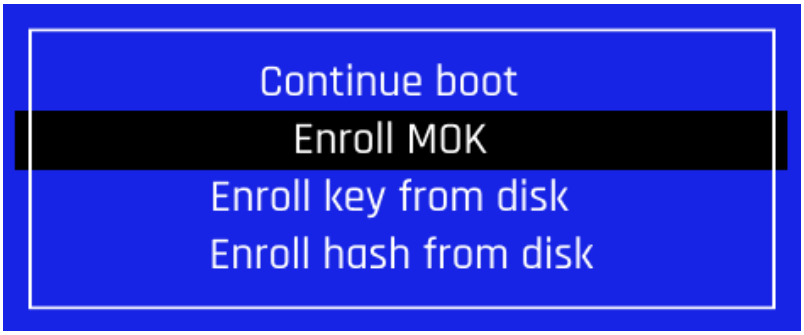
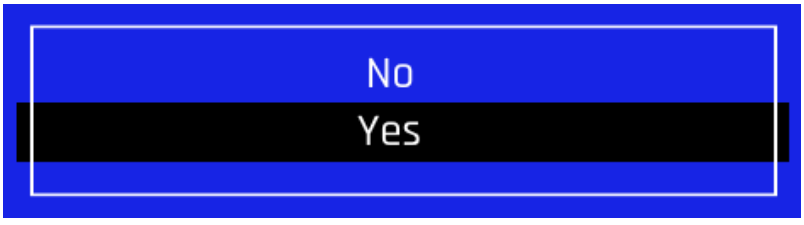


Figure 14 – Secure boot configuration password

7. After the installation is completed, reboot machine.
8. After computer reboot, the following dialog might be displayed on the screen. Choose "Enroll MOK" and follow the instructions shown below:

a. 

b. 

c. 

d. 

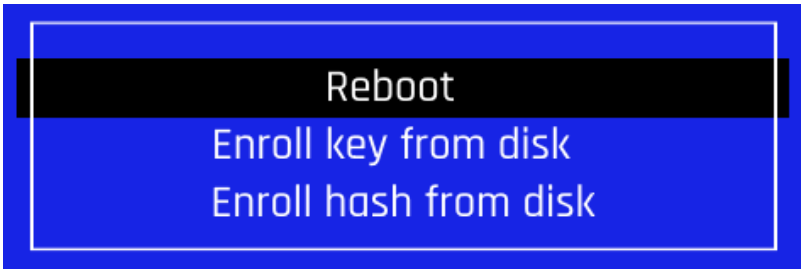
e. 

Figure 15 – Images ‘a’ to ‘e’: Enrolling MOK instructions

9. You may check the status of the driver using the following command:

```
Systemctl status kaya_driver.service
```

4.5 Building hardware driver

This section explains how to build KAYA hardware driver manually in case of a kernel update.
NOTE: This step is not a part of the installation process and should be disregarded in case the driver was added to DKMS.

1. Enter subfolder "PCI_drv_Linux" folder in the installation directory and run:

```
sh make_all.sh
```

   NOTE: This step should produce a new driver file named "predator_driver.ko"

2. Install newly built driver with the following command:

```
sudo ./ kaya_driver_install.sh
```

3. Reboot machine.

+1 (561) 689-2899
info@kayainstruments.com
www.kayainstruments.com

5 Installation Notes

5.1 How to downgrade Vision Point application from version 2019.1 to 4.4

Please follow these steps for a proper reinstallation of the Vision Point application:

1. Open the installation directory: “C:\Program Files\KAYA Instruments\Common\bin”.
2. Execute the stop service command by double-clicking the "KYService_stop.bat" file.
3. Unregister by double-clicking the "KYService_unregister.bat" file.
4. Uninstall the Vision Point application by accessing the "C:\Program Files\KAYA Instruments\Vision Point" directory and executing "unins000.exe" file.
5. Download Vision Point 4.4 version and perform a full installation.

5.2 Driver installation for Windows

KAYA's drivers are digitally signed according to the latest Microsoft driver signing policy and procedure. The digital signatures certificate can be found in KAYAKERN.sys driver properties. For a successful driver signature verification, please ensure your Windows is up to date and all the latest updates and hotfixes are installed.

5.3 Software and firmware upgrade for Windows

The new Vision Point software no longer supports devices that do not use device ID 0x1000, including all devices with any firmware below 4.0.

Devices with Hardware ID below 0x1000 won't be recognized in Windows 10; thus, Hardware ID conversion won't be available. The new software contains numerous bug fixes and improvements, so we advise you to update the software to the latest version.

Please follow the instructions below:

1. Update the device firmware to the newest version.
   NOTE: *PC power cycle is required after the firmware update (shut down, not restart).*

2. Please make sure the Hardware ID in Device manager shows the following:
   PCI\VEN_1D2A&DEV_1000&SUBSYS_YYYY1D2A&REV_07 (when the "1000" is the four-character identifier for the device, and the "YYY" is the four-character subsystem identifier).

3. Please update the software to the latest version.
   NOTE: *The order of the steps is important. It is necessary to update the device firmware to 4.0 version or higher before installing a new Vision Point software. If the software is updated first, the device won't be detected (in this case one should downgrade the software and perform steps 1-2).*

REFERENCES

Supported vision standards:

CoaXPress GEN<i>CAM

Vision Point documentation:

Vision Point

TECHNICAL SUPPORT AND PROFESSIONAL SERVICE

If you searched the documents and could not find the answers you need, contact KAYA Instruments support service:

- Create a support request on the web: support.kayainstruments.com
- Our knowledge base is available on: kb.kayainstruments.com

Visit us at kayainstruments.com for comprehensive information.

SUBMITTING A SUPPORT REQUEST

When opening a support request, please provide the following information when applicable:

For Frame Grabbers:	For Camera:	For Range Extender:
• Vision Point Diagnostic Info* • Serial number of Frame Grabber • Camera model • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP • PC motherboard model *In the Vision Point app, use menu option Help > Collect diagnostic info.	• Vision Point Diagnostic Info (or frame grabber being utilized) • Serial Number of Camera • XML dump and/or description of how the camera is being utilized • Description of issue • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP	• Range Extender Model • Serial Number of Range Extender • SFP+ module model • CoaXPress/Fiber Cable model and length • PC configuration • Operating System • Software Revision • Camera and Frame Grabber Manufacturer and Model

KAYA Instruments

Please feel free to contact our sales team for pricing, availability, documentation or customization through our e-mail or phone, we will be happy to provide assistance and consultation.

Sales Inquiries: info@kayainstruments.com

www.KAYAInstruments.com

KAYA Instruments, Inc.
2255 Glades Rd. Suite 324A
Boca Raton, FL 33431
USA

+1 561 698-2899

KAYA INSTRUMENTS

© 2024 KAYA Instruments, Inc. All rights reserved. KAYA Instruments, the KAYA Instruments Komodo logo, JetCam logo, Predator, Iron and combinations thereof are trademarks of KAYA Instruments, Inc. in the United States and/or other jurisdictions. Microsoft Windows is a registered trademark of Microsoft Corporation. Other names are for informational purposes only and may be trademarks of their respective owners. KAYA Instruments is not liable for harm or damage incurred by information contained in this document.